

University of Windsor



Department: Electrical & Computer Engineering
Advance Energy Storage Systems

Professor: Dr. Balakumar Balasingam

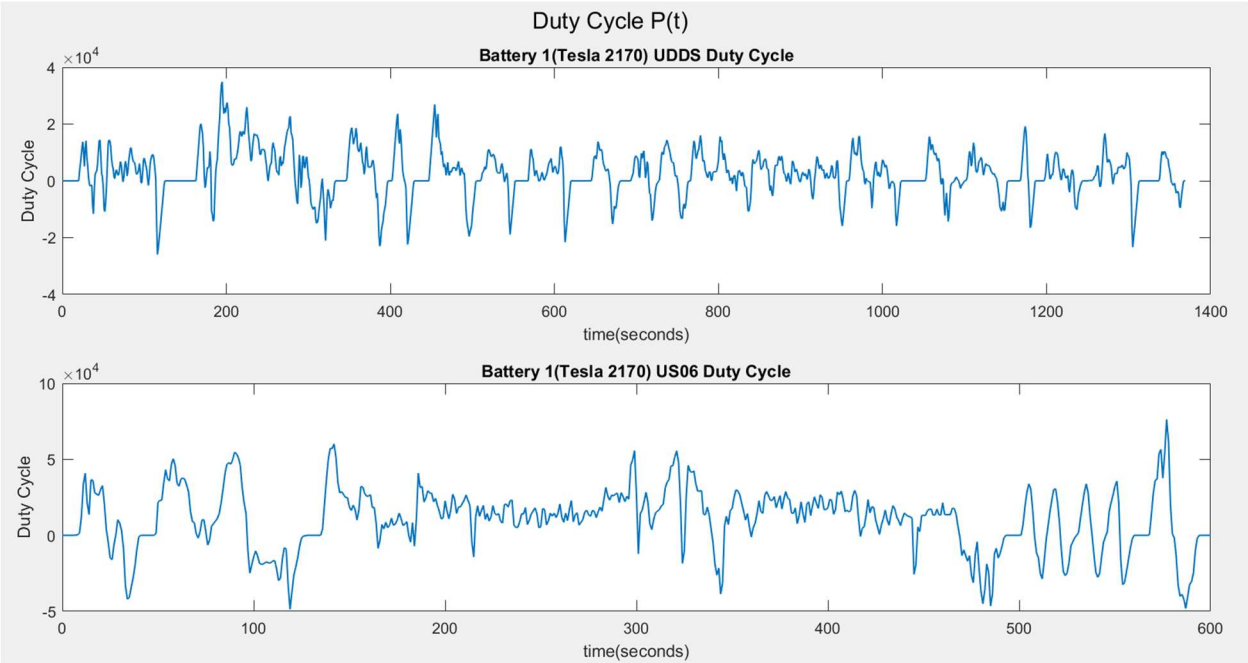
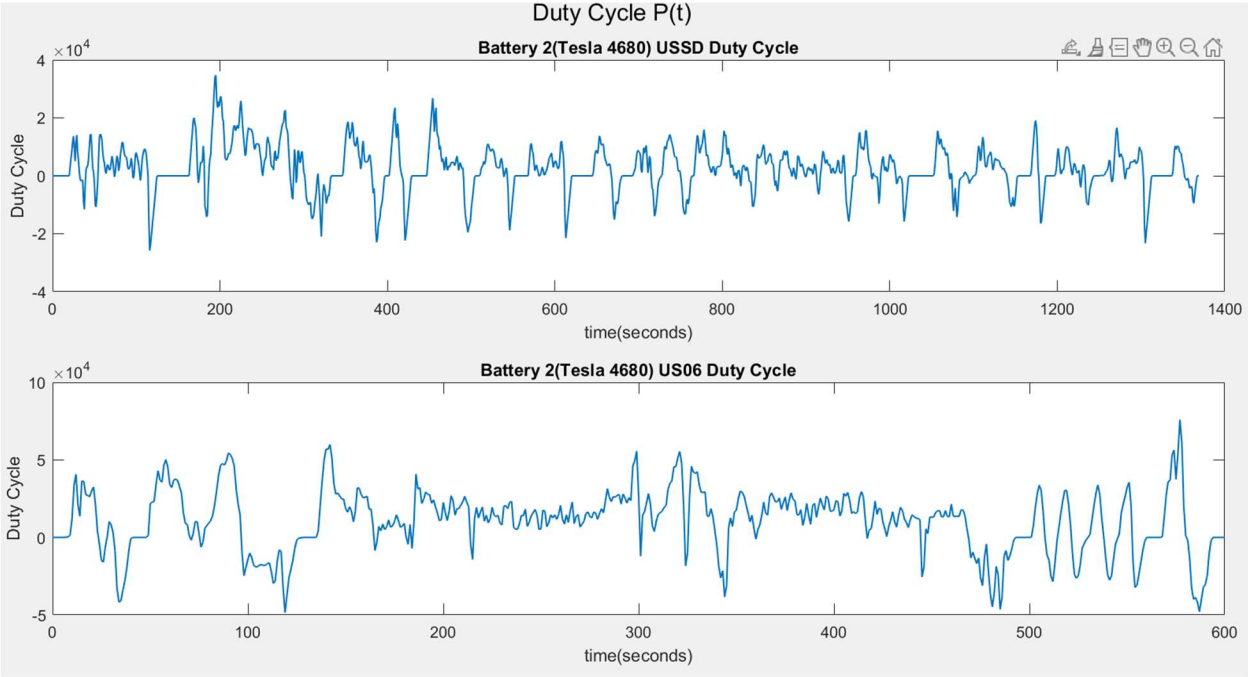
Topic:

Project 2: EV Range Estimation

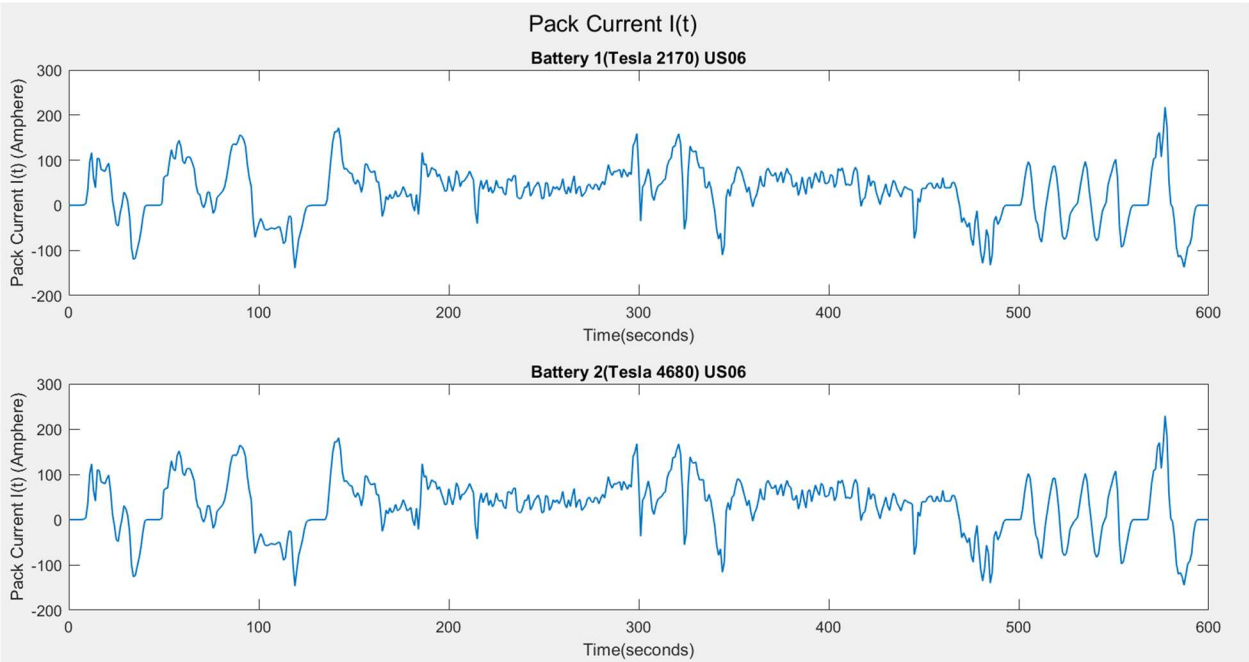
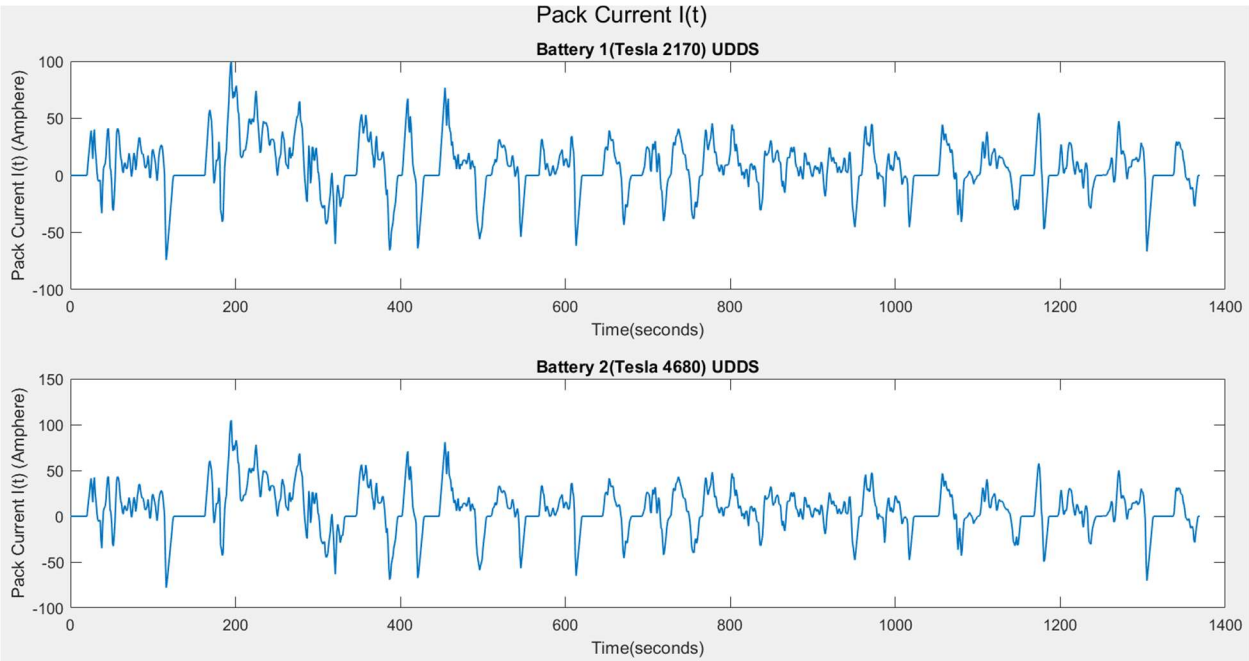
By

Jas Prajapati:110147267

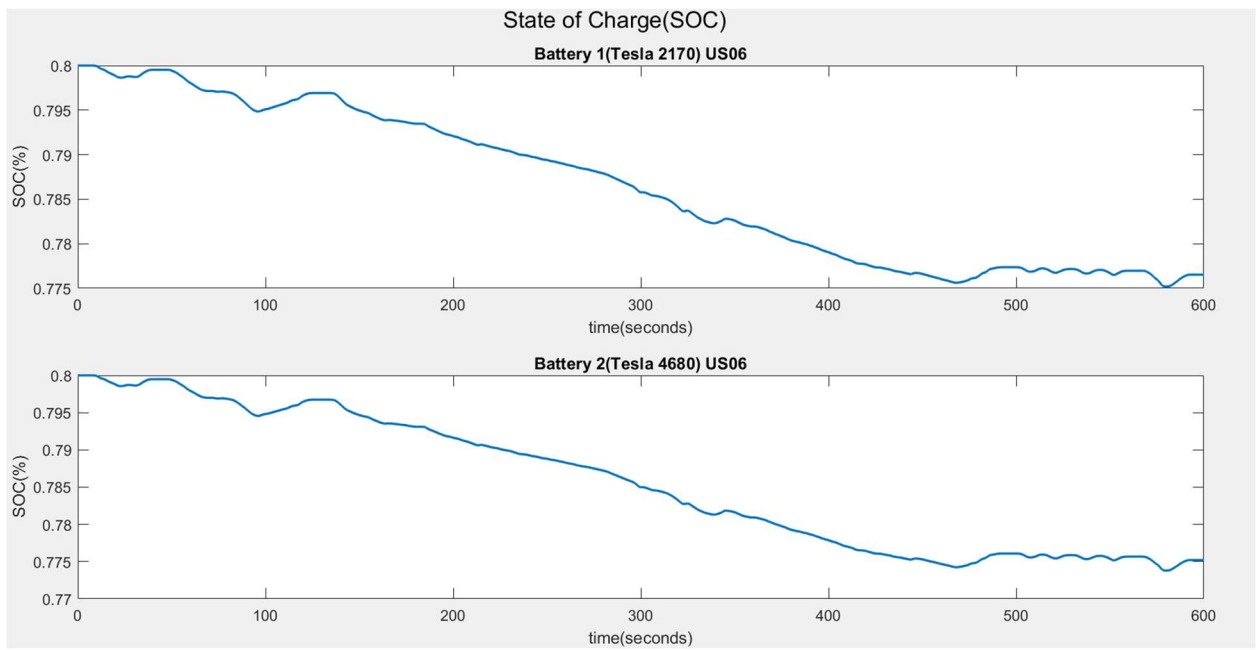
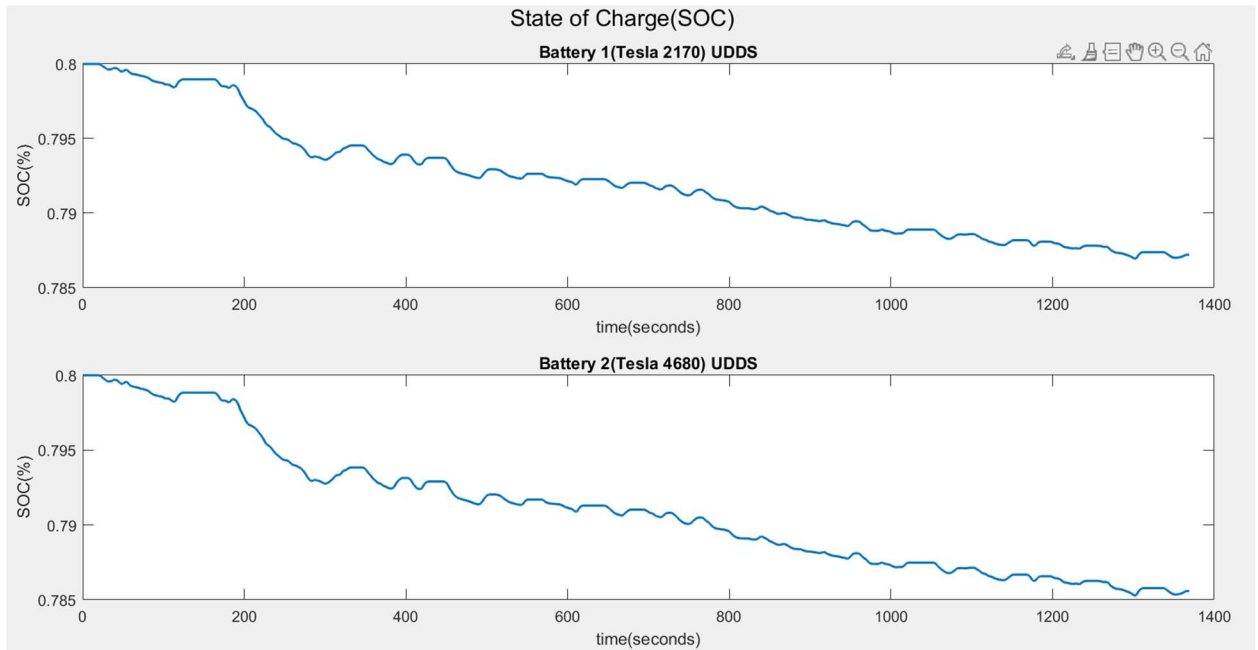
Plots for Duty Cycle P(t)



Plots for Pack Current I(t)



Plots State of Charge (SOC)



Range of Vehicle

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Command Window
Battery Pack: Tesla 2170 | Drive Cycle: UDDS
Range: 690.83 km
Energy Consumption: 107.33 Wh/km
Battery Weight: 302.94 kg

Battery Pack: Tesla 2170 | Drive Cycle: US06
Range: 439.10 km
Energy Consumption: 168.86 Wh/km
Battery Weight: 302.94 kg

Battery Pack: Tesla 4680 | Drive Cycle: UDDS
Range: 612.96 km
Energy Consumption: 106.98 Wh/km
Battery Weight: 296.42 kg

Battery Pack: Tesla 4680 | Drive Cycle: US06
Range: 389.17 km
Energy Consumption: 168.51 Wh/km
Battery Weight: 296.42 kg

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